Shape Continuation for Convex-Polytope Packing: Generalization and Lean-Verified Catalogue Certificates

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Abstract

We generalize soft-to-rigid finite-container packing from polygons and cubes to convex polyhedra and regular polytopes in dimensions four through six. Two paths are distinguished: blending with an inscribed ball, and erosion followed by dilation, which retains an actual eroded core for non-tangential solids. Continuation becomes one move in an archive search alongside symmetry-released starts, insertion, deletion, rigid perturbation, crystal cuts and nested substitution. A new Lean development proves dimension-independent certificate soundness and instantiates it on 57 actual frozen catalogue records across ten families in dimensions 3–6. The kernel checks exported raw rotation parameters, proper rotations, congruence, whole-body containment and every pairwise separation. A separate source audit binds these exact literals to the original records without changing scales or placements; parsing remains outside the kernel. The checked cohort is a subset of 402 rational exports, rather than a verification of the entire 10,398-witness archive. Independent exact-arithmetic checkers and five separately rechecked examples complement this formal layer. A historical comparison of 1,824 attempts on 40 prism-family cases illustrates why improvement frequency and magnitude lead to different search policies. The results establish constructive feasibility and an auditable method; they do not prove global optimality or universal superiority of continuation.

1 Purpose and relation to earlier work

Finite packing asks how small a prescribed container can be while holding a fixed number of congruent bodies. It combines discrete contact patterns with continuous translations and rotations. A numerical optimizer may locate an attractive configuration while leaving imperceptible overlaps; a reliable constructive bound therefore needs a separate validity argument.

Nakajima's soft-to-rigid study [1] uses gradual shape change as an alternative search path. Its controlled comparisons favor tuned rigid starts overall, while showing complementary behavior on particular polygon instances. Section 7 already credits Zarzuelo's extension to Platonic solids, generalized exact checking and repeated softening of archived packings. That downstream use motivates the present account. We do not present the broader catalogue as an independent controlled validation of the original simulator.

The earlier DOI publications document the initial nine-family implementation [2], structural analysis of its 351 constructions [3], and the associated data release [4]. The present paper concerns a later, substantially larger implementation in the shared repository [15]. Its contributions are the geometric distinction needed for non-tangential 3D solids, the supported higher-dimensional search and certification pipeline, the integration of multiple construction families into one archive, and a new, explicitly bounded Lean proof layer. The three earlier DOI objects identify the original scientific and data record; their frozen dataset must not be confused with the expanded snapshot below.

General polyhedron packing and continuous rotations predate this work [5]; Berthold et al. [6] already studied all 25 ordered Platonic-solid pairs. Rounded polyhedral particles and shape

evolution also have established uses [7]. Sequential linear programming [8], lattice packing [9], and periodic polyhedron packing [10] supply important context and seed constructions. We claim an implemented and inspectable generalization of a search-and-certification pipeline, rather than invention of these underlying geometric ideas.

The repository is an open, versioned research archive connecting independent contributions with geometry definitions, coordinate witnesses, certificates, source lineage and search code. Its literature review and historical-source audit [16, 17] connect the catalogue to prior research; its search-method study, large-population analysis and higher-dimensional notes [18, 19, 20] record the implementation's empirical development. These project research materials complement, rather than replace, citations to the original papers.

2 Packing problem and normalization

Let $P, C \subset \mathbb{R}^d$ be compact, full-dimensional convex polytopes, with fixed reference scales and the origin in their interiors. Write

$$\begin{aligned} s^*_{P\to C}(n) = \inf\{s > 0 : p_i + R_i P \subseteq sC, \quad R_i \in \mathrm{SO}(d), \\ \mathrm{int}(p_i + R_i P) \cap \mathrm{int}(p_j + R_j P) = \varnothing \quad (i \neq j)\}. \end{aligned} \tag{1}$$

A common container translation can be absorbed into the piece centers. Requiring proper rotations matters for chiral solids: a reflected enantiomer is not generally a rotated copy.

For unit-edge solids, s is container edge divided by piece edge. Catalan self-packings use a fixed canonical polar-dual shape; elongated prisms retain their specified aspect ratio. For mixed shapes, equal numerical edge ratios need not imply equal density. With compatible reference volumes,

$$\eta = \frac{n \operatorname{vol}_d(P)}{s^d \operatorname{vol}_d(C)}, \qquad s \geq \left(\frac{n \operatorname{vol}_d(P)}{\operatorname{vol}_d(C)}\right)^{1/d}. \tag{2}$$

The exponent changes with dimension; $n^{1/d}$ alone is appropriate only when the reference volumes agree. Defining $\sigma = s[\operatorname{vol}_d(C)/\operatorname{vol}_d(P)]^{1/d}$ gives $\eta = n/\sigma^d$.

Native higher-dimensional coordinates need an additional conversion. If P_0, C_0 have canonical edge lengths e_P, e_C, let $\mu = e_P/e_C$. A record at edge ratio s places copies of P_0 in

$$c_C + s\mu C_0. \tag{3}$$

Consequently raw canonical volumes give $\eta = n \operatorname{vol}_d(P_0)/[(s\mu)^d \operatorname{vol}_d(C_0)]$. Omitting μ can reverse comparisons. Likewise, comparisons with circumradius-normalized literature require the corresponding edge conversion.

3 Two continuation paths

Let $\mathbb{B}_d$ be the closed unit Euclidean ball and assume $\rho\mathbb{B}_d \subseteq P$, $\rho > 0$.

3.1 Homothetic-core blending

For $0 \leq \tau \leq 1$, set

$$H_\tau = (1-\tau)P \oplus \tau\rho\mathbb{B}_d. \tag{4}$$

Here $\oplus$ denotes Minkowski addition. The endpoints are $H_1 = \rho\mathbb{B}_d$ and $H_0 = P$. This formula is dimension-independent and does not require all facets to touch the inscribed ball.

Proposition 1. *The family satisfies $H_\tau \subseteq P$. If $0 \leq \tau_1 \leq \tau_2 \leq 1$, then $H_{\tau_2} \subseteq H_{\tau_1}$.*

Proof. Every point of H_τ is a convex combination of a point of P and a point of $\rho\mathbb{B}_d \subseteq P$. For a unit direction u, its support function is

$$h_{H_\tau}(u) = (1-\tau)h_P(u) + \tau\rho.$$

Since $h_P(u) \geq \rho$, this expression decreases with τ. Support-function comparison gives the claimed inclusion. □

Containment by convex blending is formalized in Lean as `convex_blend_subset`. The support-function nesting argument above is a mathematical proof in the manuscript, not a claim that the entire proposition's support-function formulation has been formalized.

3.2 Actual eroded cores for non-tangential solids

For a radius $r \geq 0$, define the erosion and opening

$$E_r = P \ominus r\mathbb{B}_d = \{x : x + r\mathbb{B}_d \subseteq P\}, \qquad O_r = E_r \oplus r\mathbb{B}_d. \tag{5}$$

If $P = \{x : a_f \cdot x \leq b_f,\ f = 1, \ldots, F\}$ with unit outward normals, then

$$E_r = \{x : a_f \cdot x \leq b_f - r,\ f = 1, \ldots, F\}. \tag{6}$$

This uses the actual supporting planes. For general P, it does *not* imply $h_{E_r}(u) = h_P(u) - r$ in every direction: vertices and active facets can change.

Proposition 2. *For $0 \leq r \leq R$, $O_R \subseteq O_r \subseteq P$, and $O_0 = P$.*

Proof. By definition, $y + z \in P$ whenever $y \in E_r$ and $z \in r\mathbb{B}_d$, proving $O_r \subseteq P$. If $R > 0$ and $x = y + Ru$ with $y \in E_R$, $\|u\| \leq 1$, write $x = [y + (R-r)u] + ru$. For $\|v\| \leq 1$, the point $y + (R-r)u + rv$ belongs to $y + R\mathbb{B}_d \subseteq P$ by the triangle inequality. Thus $y + (R-r)u \in E_r$ and $x \in O_r$. The case $R = 0$ and the endpoint follow directly. □

Lean's `minkowskiOpening_subset` proves opening containment for any structuring body; `minkowskiOpening_mono` concerns inclusion of the original bodies. The latter is not a formal proof of monotonicity in the radius in Proposition 2.

For a tangential polytope with every unit-normal offset equal to ρ, equation (6) gives $E_r = (1 - r/\rho)P$ for $0 \leq r < \rho$. Hence $O_r = H_{r/\rho}$ on that interval. The distinction matters for prisms with unequal facet distances and for many Archimedean solids: uniform core scaling and true erosion are different paths even though both can stay inside P.

The generalized 3D implementation precomputes the eroded core numerically, tracking its changing combinatorics. Sampling and bisection locate transitions; this preprocessing is not an exact proof of every intermediate body. Internal search units set the inradius to $1/2$ and cap the non-tangential erosion radius at 0.49, before a possibly degenerate core. The initial body is therefore rounded but need not be a sphere. The supported regular 4D–6D polytopes are tangential and use the homothetic-core formula; the implementation is not an arbitrary-dimensional erosion engine. Final rigid certificates are independent of these intermediate approximations.

4 From one continuation to an archive search

4.1 Candidate generation and tightening

The search separates three jobs: generate a promising configuration, reduce its scale numerically, and validate a final exact witness. Soft dynamics minimize squared positive overlap and wall penalties together with inward pressure. Momentum, bounded pose updates and annealed noise

help the rounded pieces rearrange. Softness decreases to zero, followed by hard settling. Quaternion orientations are used in 3D. In dimension d, skew-matrix increments supply the $d(d-1)/2$ local rotation parameters, with numerical matrices periodically reorthonormalized.

For 3D rounded cores, contact calculations combine separating-axis penetration tests and distances between disjoint features. In higher dimensions, support-point queries and optimized separating hyperplanes avoid treating 3D edge-cross-edge axes as a complete d-dimensional test. These are search calculations; their numerical residuals are not feasibility proofs.

A legalizer spreads centers if necessary, finds candidate separating planes and refits the container including its translation. Constrained tightening then minimizes scale using vertex/wall and pair-plane constraints. Sequential linear programming uses trust regions and optional regularization; an earlier constrained nonlinear path remains available. Tightening can fail, stagnate or miss an improved basin. The program name `exact.py` refers to the geometric formulation and does not make its numerical optimization exact arithmetic.

4.2 Reusing and changing arrangements

The archive stores selected witnesses and pools of distinct useful arrangements for each problem and population. A hop slightly expands and perturbs an existing packing, optionally softens a cluster or all pieces, then hardens and tightens. This gives continuation a role beyond a new random start. Related remelting and insertion were also explored in the independent cube continuation by `@BuilderOfAgents`/tronford [13]; these moves are not claimed as originating here. An insertion move starts from $n-1$ pieces, deletion starts from $n+1$, and reinsertion removes a small set before searching for better holes. Cheap rigid perturbation followed by tightening competes with soft moves.

Deleting pieces preserves feasibility, giving

$$s^*_{P\to C}(n) \le s^*_{P\to C}(n+1). \tag{7}$$

The archive propagates useful deletions and retains their lineage. A plateau in selected bounds may result from deleting a loosely constrained piece; it is not by itself a plateau theorem for the unknown optimum.

Adaptive allocation uses measured improvements, logarithmic scale gains, elapsed time, failure streaks, adjacent populations and plateau information. Exploration and polishing use different policies. Such heuristics help distribute finite computation, but no convergence guarantee is asserted.

4.3 Symmetric starts and finite crystal cuts

In 3D, `symfresh` creates orbits under a randomly selected cyclic symmetry of the container during the soft phase, then releases the symmetry before unrestricted tightening. A structured start can reduce an unproductive search space while allowing the final packing to break the imposed symmetry. A symmetric seed and a symmetric final witness are therefore distinct observations. Higher-dimensional symmetric starts also use central inversion, reflections and suitable four-element groups, while the exported pieces must still have proper rotations.

Grid, lattice, double-lattice and prism-column arrangements give further seeds. Finite windows are cut from periodic constructions, translated and rotated relative to the container, then refitted and tightened. Prior lattice and dense-periodic work [9, 10, 11] supplies context. A high infinite-volume density does not guarantee a good small finite cut: wall compatibility and missing boundary cells can dominate. Conversely, a good finite-container packing is not evidence of a physically selected crystal or a densest infinite packing.

4.4 Nested substitution and community lineage

Suppose m copies of P fit in aQ and k copies of Q fit in bC, all in compatible units. If inner poses are $R_i x + p_i$ and outer poses are $U_j y + q_j$, scale the whole outer arrangement by a and replace each outer piece by the inner packing. The resulting poses are

$$U_j R_i x + U_j p_i + a q_j \quad \text{in } abC. \tag{8}$$

Thus any subset of the mk pieces proves

$$s^*_{P\to C}(N) \le ab, \qquad 1 \le N \le mk. \tag{9}$$

This classical inclusion principle needs dimension and normalization compatibility. Container-center translations must be handled before composing poses. Boundary contacts remain legal for the mathematical packing problem; the production exporter subsequently adds clearance and rechecks its constructed witness.

The implemented 3D sweep currently uses Platonic intermediate solids; the higher-dimensional sweep uses the supported regular polytopes. It generalizes the project's whole-body inclusion contribution by PingYou Ltd [12]; related independent contributions include honeycomb windows, prism subdivisions and face lifts [14]. These construction ideas and inherited coordinates retain their original attribution. Re-padding, exact checking or combining a witness does not transfer its construction authorship. Lean's affine transport and subset-replacement lemmas formalize ingredients of this inclusion reasoning, rather than an end-to-end proof of either composition script or all instances of (9).

Other contributions include Stian's octahedron-in-cube improvements for $n = 13, 15, 18$ [21]; konoha-ily2's exact rational cube witnesses [22]; Ryan Xu's independently published cube coordinates for $n = 34, 35, 36$ [23]; and Sam Scott's tightening audit of Haowei Lin's previously published arrangements [24]. The archive also incorporates coordinate reconstructions matching Friedman's historical $n = 13, 14$ cube bounds by `@BuilderOfAgents`/tronford [25], alongside original sources from Friedman, Walsh, Lin and Berthold et al., documented in the source audit [17]. Imported original constructions, reconstructions and later refinements remain separate provenance roles. The catalogue is the combined result of these searches and contributions, not solely the work of the two manuscript authors.

5 Exact witnesses and the scope of formal proof

5.1 Finite geometric certificate

Write V_i for the transformed vertex set of piece i and let the container be an intersection of closed halfspaces. A sufficient certificate consists of

$$a_f \cdot v \le b_f \qquad (v \in V_i,\ \text{all } i, f), \tag{10}$$

$$u_{ij} \cdot v \le \ell_{ij} < r_{ij} \le u_{ij} \cdot w \qquad (v \in V_i,\ w \in V_j,\ i \ne j). \tag{11}$$

The wall offsets incorporate the scale and center of the container. Strict margins may also be required at walls. Nonzero pair normals and a positive separating gap are implicit in (11) whenever both vertex sets are nonempty.

Theorem 1. *If (10)–(11) hold, every convex hull* $\operatorname{conv} V_i$ *lies in the container and these closed convex hulls are pairwise disjoint.*

Proof. A linear inequality valid at all vertices is valid on every convex combination. Consequently each hull satisfies every wall inequality. A common point of two hulls would have its projection at once at most ℓ_{ij} and at least r_{ij}, contradicting $\ell_{ij} < r_{ij}$. □

This is a dimension-independent statement. To conclude that these hulls represent congruent copies of the intended solid, one must additionally validate the canonical geometry and the rigid placements. The certificate theorem does not infer these facts from arbitrary coordinate lists.

5.2 Production exact-arithmetic checks

The 3D exporter records full coordinates, scale and scalar-first rational quaternions. The certifier generates or receives rational separator proposals; not every native JSON stores its separating planes. A nonzero rational quaternion q denotes the exactly proper orthogonal rotation $H(q)/\|q\|^2$. Exact canonical geometry is represented in algebraic number fields or multiquadratic extensions, with positive normalization factors where needed. The certifier checks facets and geometry and decides the final inequalities using integer-based outward-rounded enclosures of exact quantities. Floating-point routines may propose a separator but cannot turn a failing exact inequality into a successful one.

Higher-dimensional records use rational skew-symmetric S and a diagonal sign matrix D with determinant $+1$, representing

$$R = D(I - S)^{-1}(I + S). \tag{12}$$

Over the reals, $I-S$ is invertible and the Cayley factor is proper orthogonal; these exact identities are used by the production representation. The native polytope, facet list, edge conversion μ, containment and separation are independently checked. Circumscribed-ball tests can certify obviously distant pairs or whole-piece containment; otherwise explicit vertices and hyperplanes are used.

The certified scale is the full `s_full` value, not the truncated table display and not the fitted zero-clearance contact estimate. Closed-form expressions stored as conjectures remain conjectures. Exact feasibility yields an upper bound on s^*, not a matching lower bound or a global optimality proof.

5.3 New Lean development

Formal verification has a substantial packing precedent: Flyspeck verified the Kepler conjecture for dense sphere packings using HOL Light and Isabelle [26]. The present development addresses feasibility of explicit finite convex-polytope configurations. It uses Lean [27] 4.31.0 and a pinned mathlib revision [28]. It defines a vertex body as the real convex hull of a finite vertex list and a container as a finite intersection of linear halfspaces. The proof interface permits heterogeneous vertex counts and any real vector space, including $\mathbb{R}^d$ for every finite d.

The original two-cube interface test remains as a regression example. The catalogue extension below instead uses archived placements and their recorded full scales.

5.4 Instantiated catalogue certificates

A proof-producing exporter targets 402 rational records from the same frozen snapshot: 99 cube-in-cube records with $2 \le n \le 100$, and 303 records in nine 4D–6D families. The completed, axiom-audited cohort contains 30 cube records and 27 higher-dimensional records, comprising 653 rigid pieces and 9,783 unordered piece pairs. The supplement lists every completed record, source path, construction credit, immutable JSON hash, generated Lean source hash, compiler result and theorem axioms. Exported, pending, failed or timed-out records are not counted as completed proofs.

For each cube placement, the generated data retain the exact raw quaternion q, prove $\|q\|^2 > 0$, and check equality of the supplied rational matrix with the normalized quaternion formula. Orthogonality and determinant one are checked exactly. For higher-dimensional placements, an explicit inverse of $I - S$ is checked against the raw skew matrix and diagonal signs; a

Lean declaration	Kernel-checked meaning
`vertex_tests_imply_containment`	Vertex/wall inequalities imply whole-hull containment.
`separating_vertex_tests_imply_disjoint`	Strict projection gaps separate the entire hulls.
`finite_certificate_sound`	Finite wall and pair tests imply containment and pairwise disjointness.
`affine_placement_of_vertex_body`	Affine placement commutes with taking the vertex hull.
`transported_packing`, `subpacking_sound`, `nested_packing_sound`	Injective transport, subset replacement and disjoint child packings inside disjoint parents preserve a packing.
`minkowskiOpening_subset`, `convex_blend_subset`	The two continuation constructions remain inside the target body under the stated hypotheses.
`Rational.checkPlaced_sound`	Exact rational tests imply rigidly placed whole-body containment and separation.
`Cayley.check_sound`	Checked raw skew/sign parameters give the stated Cayley rotation with determinant one.
`Barycentric.check_sound`	Rational convex combinations prove containment in the actual container vertex hull.
`Catalogue.*.packing`	Per-record feasibility theorems for the completed catalogue cohort.

Table 1: Formal principles and their catalogue instantiations. The full website archive remains outside the completed verification cohort.

Dimension	Families	Completed	Export targets
3	Cube in cube	30	99
4	Tesseract, 16-cell, 24-cell in themselves; 16-cell/24-cell both ways	15	195
5	Hypercube and orthoplex in themselves	6	62
6	Hypercube and orthoplex in themselves	6	46
	Total	57	402

Table 2: Completed catalogue feasibility proofs, distinct from all successfully exported proposed certificates. Per-record details and piece counts are in the hash-matched coverage manifest.

generic theorem identifies the resulting matrix with $D(I-S)^{-1}(I+S)$ and proves determinant one. Orthogonality implies preservation of squared Euclidean distance, so the theorem concerns congruent pieces rather than independently displaced vertex clouds.

Cube containment is linked to the ordinary unit cube and the side-`s_full` container. In higher dimensions each placed vertex carries nonnegative rational barycentric weights summing to one in the scaled canonical container vertices. Lean checks all coordinate equalities, then convexity extends containment to the entire piece. The actual 4D, 5D and 6D hypercube templates are additionally identified with coordinate unit boxes. For the orthoplex and 24-cell families, the formal conclusion concerns their supplied canonical coordinate hulls; a separate regular-polytope classification theorem is not claimed.

Every unordered pair receives an explicit rational separating plane, including pairs for which the production checker used a distant-pair shortcut. Strict projection gaps prove disjoint closed bodies. Cube support formulas reduce repeated corner enumeration, and bounded computations are assembled from separate source, Gram, wall, pair and barycentric proofs. All numerical decisions use `decide+kernel`; no compiled decision oracle is trusted.

The exporter preserves production numeric semantics: JSON integer literals remain exact integers, decimal or exponent literals are decoded to binary64 and then converted to their exact rational values, and quoted rational strings retain their exact values. No scale, centre or orientation is perturbed to make a proof pass. All 402 source bindings have a separate exact Python audit, which is not counted as Lean verification. Parsing, source selection and transcription remain outside the kernel, with immutable inputs and byte hashes making that boundary inspectable.

The formal source uses no `sorry`, `admit`, `native_decide` or custom axioms. Build receipts report the standard logical dependencies `propext`, `Classical.choice` and `Quot.sound`. The optimizer, intermediate core preprocessing, production interval implementation, remaining algebraic-coordinate geometry and uninstantiated compositions are not verified. The 10,398 archive certification flags below therefore remain distinct from the 57 completed catalogue proofs.

6 Frozen computational evidence

6.1 Coverage and five inspectable witnesses

The computational snapshot is repository commit `1abc63218db34cea95802aeb3b6d4a6b6fe1c61b`, with site data built on 15 October 2026 at 23:21 and retrieved on 16 October. Table 3 is a coverage statement about the selected catalogue. It includes inherited, community, structured and numerically searched constructions, not 10,404 new discoveries or independent continuation successes.

Family	Problems	Population	Slots	Coordinate witnesses
3D Platonic pairs	25	2–100	2,475	2,469
3D Archimedean self-packings	13	2–100	1,287	1,287
3D Catalan self-packings	13	2–100	1,287	1,287
3D prism-related problems	35	2–100	3,465	3,465
4D regular pairs	36	2–40	1,404	1,404
5D regular pairs	9	2–32	279	279
6D regular pairs	9	2–24	207	207
Total	140		10,404	10,398

Table 3: Frozen selected-catalogue coverage. All coordinate entries are marked exact-certified in the snapshot. The six remaining entries are source-only published octahedron-in-cube bounds; their coordinates are not supplied as certified witnesses here.

Five full native witnesses and their geometry are frozen with the paper (Table 4). Each was rerun through the appropriate production exact checker for this manuscript and passed. This is a recheck of the selected examples, not a fresh recertification of the entire archive. All five selected constructions are credited to Alejandro Zarzuelo Urdiales in their native provenance; other catalogue entries retain their own authors.

Pieces and container	d	n	Certified full s	Recorded move
Pentagonal prisms in themselves	3	12	2.6180343063072	symmetric fresh
Truncated octahedra in themselves	3	8	2.5000001428871	symmetric fresh
Pentagonal hexecontahedra in themselves	3	4	2.2051534370001	insertion
5-cells in a tesseract	4	4	0.8896812614235	perturbation
5-cells in a tesseract	4	8	0.9771978857222	composition

Table 4: Illustrative upper bounds, with complete coordinates and certificate logs in the supplement. A recorded final move is provenance, not a controlled attribution of the entire outcome to that move.

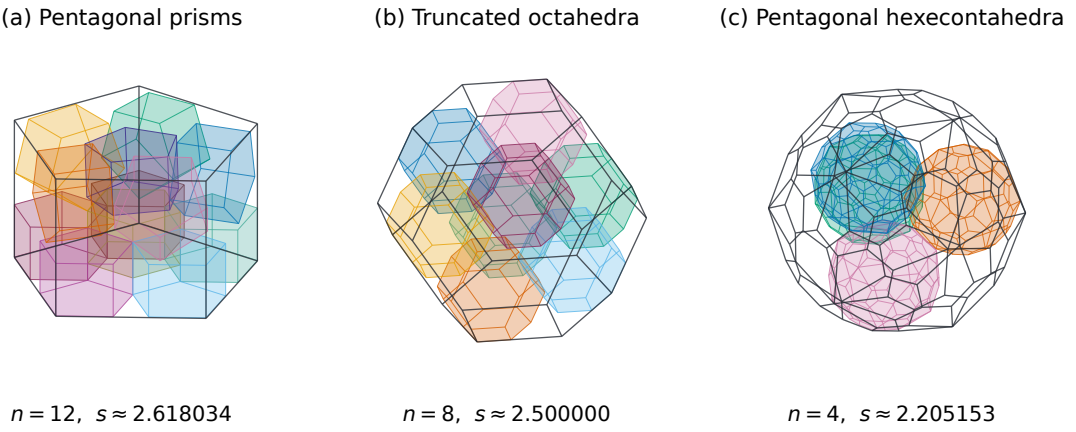


Figure 1: Three exact-rechecked 3D witnesses from Table 4: 12 pentagonal prisms, eight truncated octahedra and four pentagonal hexecontahedra in homothetic containers. Transparent bodies and wireframe containers show interior structure. Colors identify pieces; they do not assert distinct orientation classes. The rendering uses the native positions and rotations without spreading or altering the construction.

The prism and truncated-octahedron cases illustrate symmetric starts on non-tangential geometry. Their apparent contact scales $(3+\sqrt{5})/2$ and $5/2$ are recorded conjectures, distinct from the slightly larger certified scales. The chiral Catalan example has 92 vertices and 60 facets per piece and requires 22,080 vertex/facet inequalities plus six separating-plane checks. Proper-rotation treatment is essential to its interpretation. The four-dimensional examples exercise native SO(4) poses and the conversion $\mu = 2\sqrt{2}$. The eight-piece case is a nested inclusion through a 16-cell; its recorded expression $(5\sqrt{2}-\sqrt{10})/4$ is also only a conjectured contact limit.

The earlier patterns study [3] analyzes scale jumps and orientational domains in its smaller freeze. In the expanded setting, deletion, symmetry release and nested substitution provide specific mechanisms that can generate plateaus, orientation families and unexpectedly competitive mixed pairs. They offer explanations to test rather than proofs that the observed best constructions are optimal or that all their patterns persist to larger populations.

6.2 A reproducible historical comparison of search moves

We recompute the archived Round-D comparison from its six raw JSONL shards. It contains 40 cases across all 20 prism/antiprism problems then supported, with populations from 6 to 28, 1,824 valid attempts and 40,751.5 summed per-attempt elapsed seconds. The script measures elapsed wall-clock time per attempt; summing these intervals across workers does not measure processor utilization or total campaign duration. These are historical internal runs, not a new randomized or independently executed experiment. Every gain is measured against the reference scale stored in that attempt's log.

For attempts a of one method, define positive logarithmic gain and gain per aggregate worker-hour by

$$g_a = \max\{0, \log(s_{\mathrm{ref},a}/s_a)\}, \qquad G = \frac{\sum_a g_a}{\sum_a t_a/3600}. \tag{13}$$

The table reports wins using the original log's success flag, rather than silently replacing its decision rule with another tolerance. These are not matched-start comparisons: reference states and hop pools were read from an evolving archive.

Move	Attempts	Wins	Win rate	Mean seconds	G
Fresh	607	71	11.70%	25.39	0.2922
Hop	608	200	32.89%	16.60	0.1651
Symmetric fresh	609	86	14.12%	25.04	0.5446

Table 5: Recomputed Round-D outcomes. The largest improvement frequency and the largest summed improvement per aggregate worker-hour belong to different moves. Timing and archive dependence limit generalization beyond this cohort.

Symmetric fresh starts give 3.30 times hop's and 1.86 times fresh's measured G, whereas hops improve the reference more often. Among 31 cases improved by more than 0.1%, the lowest result comes from symmetric fresh in 21, fresh in five and hop in five. The evidence supports keeping distinct search mechanisms available; it does not estimate a causal benefit of temporary softening, demonstrate a fixed-budget portfolio advantage, or establish a universal speedup. There are no independent reruns or uncertainty intervals here, and Elapsed-time observations may depend on implementation, machine load and the selected historical cases.

7 Limitations and testable next steps

The generalized method changes the set of configurations that can be explored and certified, but its numerical search remains local and stochastic. Catalogue completeness over a displayed range is not exhaustive optimization. Source-only bounds and certified coordinate witnesses are distinct evidence classes. A rigorous upper bound can be worse than an uncertified contact estimate while still being the scientifically stronger feasibility statement.

A direct continuation ablation should start from the same archived hard packings, allocate the same measured budget to rigid perturbation and softening/rehardening, use identical tightening and exact acceptance rules, and reserve unseen problems for evaluation. Symmetric and crystal seeds should be compared at equal total cost, including seed construction. Outcomes should include the best certified scale, time-to-target and run-level failures, with uncertainty estimated across independently chosen seeds. The present archive and Round-D reanalysis do not replace that experiment.

Several structural hypotheses are now testable. A plateau may arise from a removable, weakly constrained piece; a sharp gain may mark a change in boundary-compatible layers or orientation domains. Symmetric starts may help when good solutions retain approximate orbits but require a small final symmetry break. Non-tangential opening may preserve useful face

contacts better than uniform core scaling. Nested substitution may be especially effective when one-piece inclusions are efficient and the intermediate solid packs well. Each prediction requires comparisons against alternative constructions and compatible normalizations; none follows from visualization alone.

The Lean layer now certifies selected actual catalogue witnesses, including their exact rigid-map representations. Extending completed coverage requires additional kernel builds, algebraic-coordinate support, regular-polytope identification beyond boxes, and a verified parser if source decoding is to enter the formal trust boundary. The present evidence is 57 kernel-checked frozen catalogue instances, generic soundness theorems, and separately audited computational components; it is not a proof of the entire live archive or its search software.

8 Reproducibility and methodological tools

The supplement contains the LaTeX source, figure-generation code, the Lean library and per-record proof sources, immutable rational-cohort inputs, proof-producing exporters, per-record build and axiom receipts, five independent production-checker rerun logs, a compressed complete catalogue snapshot, a per-case provenance CSV, the historical benchmark shards and aggregation code, and source/hash manifests. The implementation is available at https://github.com/alejandrozu/platonic-packing; the interactive catalogue is https://polyhedrainpolyhedra.wiki/. The repository commit named above freezes the computational implementation and data used here; the accompanying manuscript commit separately timestamps this document and its formal development.

The earlier DOI releases [2, 3, 4] remain citations to their original scope and freeze. They are not retroactive identifiers for the current 140-problem archive. Record links, `s_full`, geometry normalization and construction lineage should be retained when reproducing or extending an example.

Software tools include Python, NumPy/SciPy and numerical optimizers, JavaScript dynamics and visualization, Lean/mathlib, and LaTeX. AI coding and drafting tools, including Claude and Codex, assisted implementation, analysis and manuscript preparation under human direction. Generated suggestions are not evidence of correctness; the mathematical arguments, exact checker outputs and Lean build artifacts identify the independently inspectable support for each claim. The recorded construction credit remains distinct from software assistance.

References

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